



☒ SIH26036 — Complete System Documentation

Project Title: Online Verification System for Weighing and Measuring Instruments

Problem Statement ID: SIH26036

Ministry: Ministry of Consumer Affairs, Food & Public Distribution (Department of Consumer Affairs)

Legal Framework: Legal Metrology Act, 2009 + Legal Metrology (General) Rules, 2011 + GATC Rules, 2013

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1. Executive Summary

1.1 Problem Statement

Development of an online verification system for weighing and measuring instruments used in trade and commerce. The system must digitize the entire lifecycle from instrument registration → verification application → inspection → certificate generation → public verification → renewal.

1.2 Objectives

- Eliminate manual, paper-based verification processes
- Provide transparency to citizens through public certificate verification
- Reduce processing time from weeks to days
- Ensure compliance with Legal Metrology Act, 2009
- Enable periodic re-verification tracking
- Prevent fake certificates through QR-based authentication

1.3 Scope

The system covers:

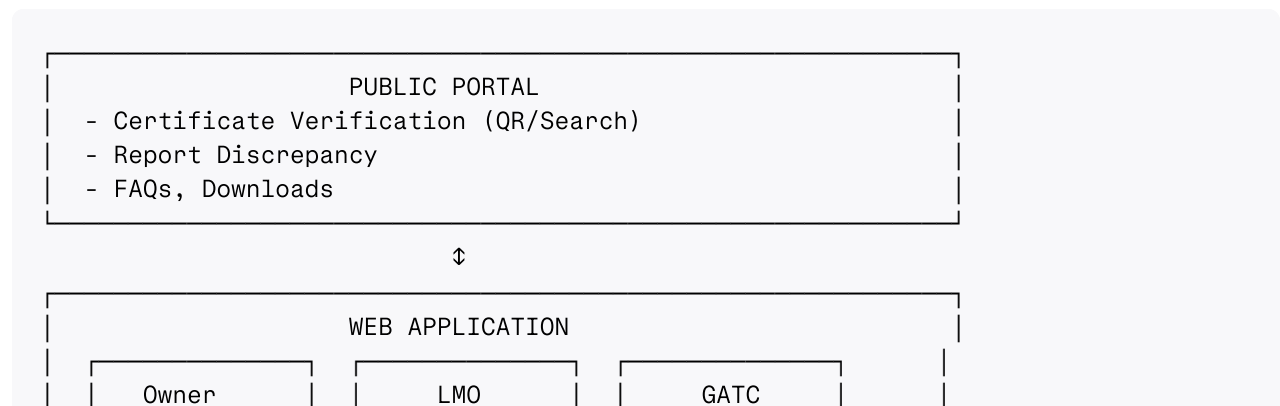
- Instrument owner registration and instrument pre-registration
- Online application submission for verification/re-verification
- LMO (Legal Metrology Officer) field inspection workflow
- GATC (Government Approved Test Centre) laboratory testing workflow
- Digital certificate generation in Schedule VIII format
- Public portal for certificate verification via QR code/search
- Renewal/re-verification lifecycle management
- Role-based dashboards for all stakeholders

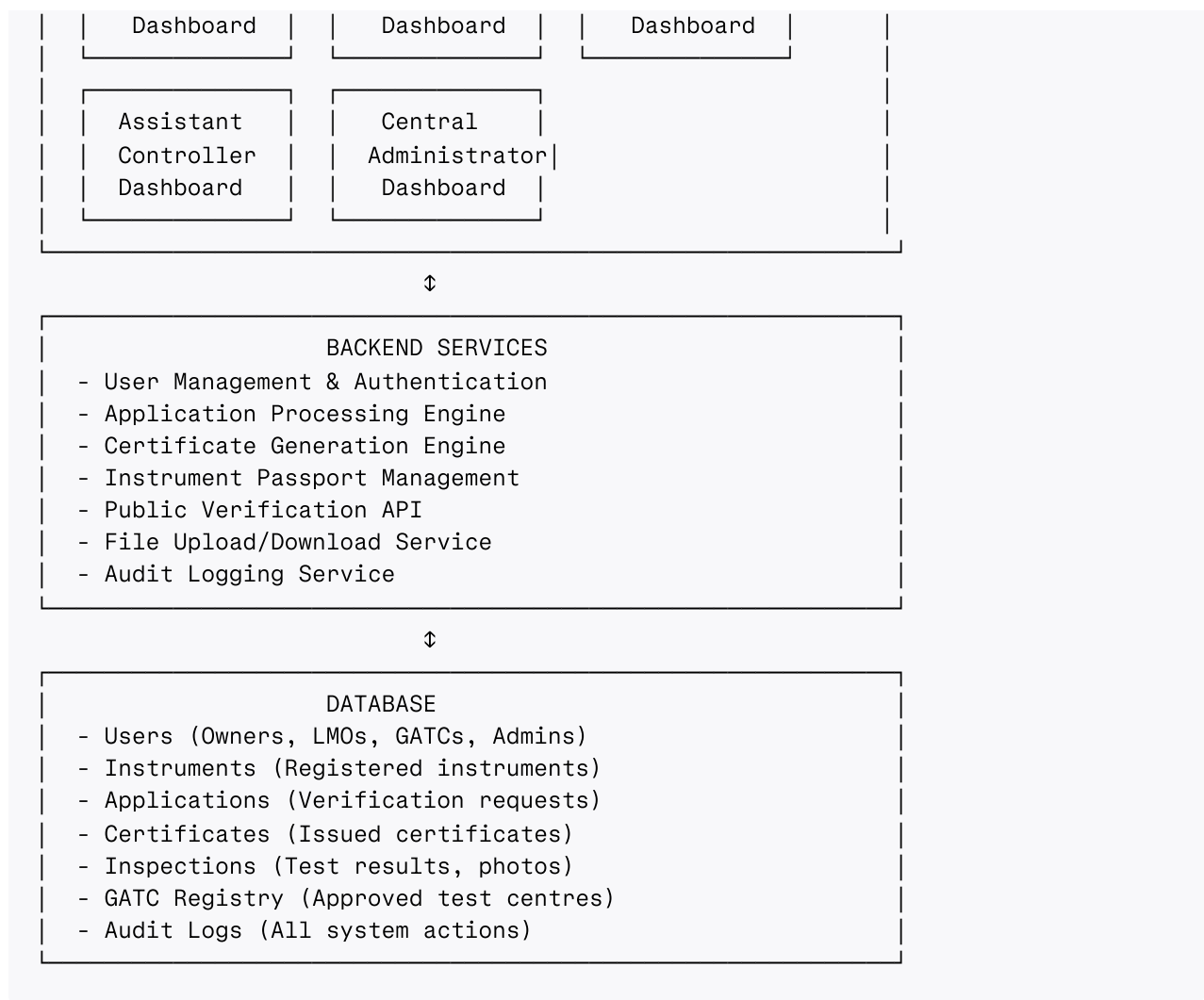
1.4 Out of Scope (for MVP)

- Online payment gateway integration (manual payment for now)
- SMS/Email notification system (manual communication for now)
- Digital signature integration (text-only approval for now)
- Advanced analytics and MIS reporting (basic reports only)

2. System Overview

2.1 System Architecture (Logical)





2.2 Key Features

- **Multi-role access:** 6 distinct user roles with separate dashboards
- **Instrument Passport:** Complete lifecycle history for each instrument
- **QR-based verification:** Public can verify certificates instantly
- **Automated routing:** System routes applications to LMO or GATC based on instrument type
- **Schedule VIII compliance:** Certificates match government format exactly
- **Validity tracking:** Color-coded expiry status for easy monitoring
- **Grievance module:** Citizens can report fake/expired certificates
- **Audit trail:** All actions logged for compliance

2.3 User Categories

1. **External Users:** Instrument Owners, Public Verifiers
2. **Field Users:** LMOs (Legal Metrology Officers)
3. **Laboratory Users:** GATCs (Government Approved Test Centres)
4. **Administrative Users:** Assistant Controllers, Central Administrators

3. User Roles & Permissions

3.1 Role Matrix

Feature	Instrument Owner	LMO	GATC	Assistant Controller	Central Admin	Public Verifier
Register account	✓	✗	✗	✗	✗	✗
Register instruments	✓	✗	✗	✗	✗	✗
Submit verification application	✓	✗	✗	✗	✗	✗
Upload payment receipt	✓	✗	✗	✗	✗	✗
Track application status	✓	✓ (assigned)	✓ (assigned)	✓ (district)	✓ (all)	✗
Download certificates	✓ (own)	✗	✗	✗	✗	✗
View instrument passport	✓ (own)	✓ (all)	✓ (tested)	✓ (district)	✓ (all)	✓ (summary)
Scrutinize applications	✗	✓ (assigned)	✗	✓ (district)	✗	✗
Schedule inspections	✗	✓ (own)	✓ (own)	✗	✗	✗
Conduct inspection	✗	✓ (field)	✓ (lab)	✗	✗	✗
Upload inspection report	✗	✓ (own)	✓ (own)	✗	✗	✗
Upload test report (GATC)	✗	✗	✓ (own)	✗	✗	✗
Issue certificates	✗	✓ (approved)	✗ (recommends only)	✗	✗	✗
Approve GATC reports	✗	✓	✗	✗	✗	✗
View district MIS	✗	✗	✗	✓ (own district)	✓ (all)	✗
Manage users (LMO/GATC)	✗	✗	✗	✓ (create LMOs)	✓ (create GATCs, ACs)	✗
Configure GATC instrument list	✗	✗	✗	✗	✓	✗
Verify certificates (public)	✓	✓	✓	✓	✓	✓

Feature	Instrument Owner	LMO	GATC	Assistant Controller	Central Admin	Public Verifier
Report discrepancy	✓	✓	✓	✓	✓	✓
Approve seizures	✗	✗ (proposes)	✗	✓	✗	✗
Handle appeals	✗	✗	✗	✓	✗	✗

3.2 Detailed Role Definitions

3.2.1 Instrument Owner / Applicant

Who: Business owners, traders, manufacturers, dealers, transporters, fuel station operators, hospitals, factories, pharma companies using weighing/measuring instruments for trade.

Permissions:

- Register account on portal
- Register instruments (pre-registration)
- Submit verification/re-verification applications
- Upload payment receipts (manual payment)
- Track application status
- Download issued certificates
- View instrument passport (own instruments)
- Apply for re-verification (renewal)
- Decommission instruments
- Report discrepancies (whistleblower)

Restrictions:

- Cannot view other owners' instruments/applications
- Cannot modify certificates
- Cannot approve/reject applications
- Cannot access LMO/GATC/admin dashboards

3.2.2 Legal Metrology Officer (LMO)

Who: Government Inspector / Assistant Controller at district level (field officer) responsible for verification and enforcement.

Permissions:

- View assigned applications (based on jurisdiction)

- Scrutinize applications (accept/request clarification/reject)
- Schedule field inspections
- Conduct on-site verification tests
- Upload inspection reports with photos
- Affix physical seals (log seal numbers in system)
- Issue Schedule VIII certificates (digital approval)
- View instrument passport (all instruments in jurisdiction)
- Propose instrument seizures (under Section 15)
- View pendency reports (own applications)
- Handle grievances (assigned complaints)

Restrictions:

- Can only access applications in assigned jurisdiction (pincode range)
- Cannot approve own seizures (requires Assistant Controller)
- Cannot modify GATC test reports
- Cannot create/delete user accounts
- Cannot access other districts' data

3.2.3 Government Approved Test Centre (GATC)

Who: NABL-accredited private/government laboratory authorized by Central Government to test high-precision instruments.

Permissions:

- View assigned testing requests (selected by applicants)
- Schedule laboratory appointments
- Receive instruments (log receipt in system)
- Conduct laboratory testing (controlled environment)
- Upload test reports with raw data, photos, calibration certificates
- Recommend certification or rejection to LMO
- View instrument passport (instruments tested by them)
- Manage lab capacity (available slots)
- View accreditation status (validity, categories approved)

Restrictions:

- Cannot issue certificates (only recommends to LMO)
- Can only test instruments within approved categories (per GATC accreditation)
- Cannot access field inspection data (LMO inspections)

- Cannot modify fee structures
- Cannot approve seizures

3.2.4 Assistant Controller (District)

Who: District-level supervisor of LMOs, responsible for oversight, grievance redressal, and enforcement approval.

Permissions:

- View all applications in assigned district(s)
- Review LMO actions (scrutiny decisions, inspection reports)
- Handle appeals (rejected applications, penalty disputes)
- Approve/reject seizure proposals (from LMOs)
- Create LMO accounts (under their district)
- View district MIS (verification coverage, pendency, compliance rate)
- Monitor LMO performance (inspections completed, average time to certify)
- Handle grievances (escalated from LMOs)
- Generate district reports (export to PDF/Excel)

Restrictions:

- Can only access assigned district(s)
- Cannot issue certificates (LMO function)
- Cannot conduct inspections (LMO function)
- Cannot create GATC accounts (Central Admin function)
- Cannot modify national configurations (GATC list, validity periods)

3.2.5 Central Administrator (DoCA, GoI)

Who: Department of Consumer Affairs, Government of India officials managing national portal and policy.

Permissions:

- Configure GATC-approved instrument list (Schedule I/II from GATC Rules)
- Create Assistant Controller accounts (all states/districts)
- Create GATC accounts (all accredited labs nationwide)
- View national dashboard (all states, districts)
- Upload policy documents/circulars (PDF)
- Maintain GATC registry (accreditation numbers, categories, validity)

- View public verification statistics (QR scans, discrepancy reports)
- Deactivate any user account (LMO, GATC, AC)
- Configure master data (states, districts, pincodes, instrument types)

Restrictions:

- Cannot issue certificates (LMO function)
- Cannot conduct inspections (LMO/GATC function)
- Cannot modify issued certificates (immutable once issued)
- Cannot access payment data (not integrated in MVP)

3.2.6 Public Verifier

Who: Any citizen, consumer, competitor, or enforcement agency verifying certificate authenticity.

Permissions:

- Search certificates by certificate number
- Search certificates by instrument serial number
- Search certificates by owner/business name
- Scan QR codes on certificates/instruments
- View certificate summary (validity, instrument details, owner details)
- Report discrepancies (fake/expired certificates, seal damage)
- Download sample formats (Schedule VIII, fee schedule, user manuals)
- View FAQs and help content

Restrictions:

- No login required (completely public)
- Cannot view full instrument passport (summary only)
- Cannot modify any data
- Cannot access application status
- Cannot contact owners directly (report goes to LMO)

4. Module-Wise Functional Requirements

4.1 User Management Module

4.1.1 Registration (Instrument Owner)

Requirements:

- System shall provide registration form with fields:
 - Business Type (dropdown: Manufacturer/Dealer/Repairer/Importer/Packer/User)
 - Business Name (text, required)
 - GSTIN (text, 15 characters, format validation)
 - PAN (text, 10 characters, format validation)
 - Aadhaar Number (text, 12 digits, format validation)
 - Mobile Number (numeric, 10 digits)
 - Email ID (email format)
 - Business Address (text + pincode lookup)
 - Instrument Installation Address (can add multiple addresses)
 - Document uploads (GST certificate, trade license, address proof; PDF/JPG, max 5MB each)
- System shall validate all fields (format checks)
- System shall generate unique User ID (format: LM-USER-YYYY-XXXXX)
- System shall display temporary password (user must change on first login)
- System shall set account status to "Active"
- System shall force password change on first login

4.1.2 Login

Requirements:

- System shall provide login page with User ID and Password fields
- System shall validate credentials against database
- System shall redirect to role-specific dashboard upon successful login
- System shall display error message for invalid credentials
- System shall lock account after 5 consecutive failed login attempts (unlock after 30 minutes or admin reset)
- System shall log login timestamp and IP address (audit trail)

4.1.3 Password Management

Requirements:

- System shall provide "Forgot Password" functionality
- System shall allow password reset via security questions (for MVP; OTP later)
- System shall enforce password policy:
 - Minimum 8 characters
 - At least 1 uppercase letter
 - At least 1 lowercase letter
 - At least 1 number
 - At least 1 special character
- System shall not allow reuse of last 5 passwords
- System shall require password change every 90 days

4.1.4 User Account Management (Admin Functions)

Requirements:

- Assistant Controller shall be able to create LMO accounts:
 - Employee ID, badge number, designation
 - Jurisdiction (district/zone, pincode range)
 - Email ID, mobile number
 - Default password (forced reset on first login)
- Central Administrator shall be able to:
 - Create Assistant Controller accounts (assign districts)
 - Create GATC accounts (accreditation number, categories, validity)
 - View all users (filter by role, state, district, status)
 - Deactivate/reactivate user accounts
 - Reset user passwords
- System shall log all user management actions (audit trail)

4.2 Instrument Registration Module

4.2.1 Pre-Registration of Instruments

Requirements:

- System shall allow Instrument Owners to register instruments before applying for verification

- System shall provide form with fields:
 - Instrument Type (dropdown: Electronic Weighing Scale, Platform Scale, Weighbridge, Fuel Dispenser, Tank Lorry, Analytical Balance, Flow Meter, etc.)
 - Make & Model (text)
 - Serial Number (text, unique per manufacturer)
 - Capacity & Division (numeric, e.g., 100 kg / 10 g)
 - Manufacturing Year (numeric, 4 digits)
 - Type Approval Certificate (TAC) upload (PDF, required for new instruments)
 - Installation Address (dropdown from registered addresses)
 - Previous Certificate Details (if re-verification: certificate number, expiry date)
- System shall validate:
 - Serial number uniqueness (warn if duplicate, but allow if different owner)
 - TAC upload (required for new instruments; optional for re-verification)
 - Expiry date (if provided, must be in future for re-verification)
- System shall generate unique Instrument ID (format: IN-MET-YYYY-XXXXX)
- System shall set instrument status to "Registered – Awaiting Verification"
- System shall allow multiple instruments to be registered under one owner

4.2.2 Instrument Passport

Requirements:

- System shall maintain complete history for each instrument (Instrument Passport)
- Instrument Passport shall include:
 - Instrument details (type, make, model, serial number, capacity)
 - Owner details (current and historical owners if transferred)
 - All certificates issued (initial + renewals) with validity periods
 - All inspection reports (LMO field inspections + GATC lab tests)
 - All seal numbers affixed (with dates)
 - All complaints/discrepancies reported
 - All seizure records (if any)
 - Decommission status (if applicable)
- System shall allow:
 - Owner to view own instrument passport (full details)
 - LMO/GATC to view instrument passport (all instruments)
 - Public to view summary (certificate status, validity, owner name, instrument type)

- System shall update instrument passport automatically on each transaction (certificate issuance, inspection, seizure, etc.)

4.2.3 Instrument Transfer/Ownership Change

Requirements:

- System shall allow instrument ownership transfer (if business sold/transferred)
- New owner shall register account and submit transfer request
- System shall require:
 - Sale agreement/document upload
 - Previous owner approval (via dashboard)
 - New owner acceptance
- System shall update instrument passport with new owner details
- System shall retain historical owner information
- System shall require new verification application (existing certificate remains valid until expiry)

4.2.4 Instrument Decommissioning

Requirements:

- System shall allow owners to decommission instruments (stop using)
- Decommissioning form shall include:
 - Reason (dropdown: Business closure, Instrument replaced, Damaged beyond repair, Other)
 - Additional comments (text, optional)
- System shall update instrument status to "Decommissioned"
- System shall stop expiry reminders (for future module)
- System shall retain historical records (10 years minimum)
- System shall allow reactivation (treated as new verification; fresh application required)

4.3 Application Submission Module

4.3.1 New Verification Application

Requirements:

- System shall allow Instrument Owners to submit verification applications
- Application form shall include:
 - Instrument selection (from pre-registered list or add new)

- Verification Type (radio: Initial Verification / Re-verification / After Repair)
- Preferred Mode (radio: Field Inspection by LMO / GATC Laboratory Testing)
- Preferred Date (date picker; exclude weekends/holidays)
- Preferred Time Slot (dropdown: 9-11 AM, 11 AM-1 PM, 2-4 PM, 4-6 PM)
- Special Instructions (text, optional)
- Supporting documents upload (purchase invoice, repair bill, previous certificate; PDF/JPG, max 5MB each)
- System shall auto-fill instrument details if selected from pre-registered list
- System shall validate:
 - Preferred date not in past
 - Required documents uploaded
 - TAC uploaded for new instruments
- System shall generate unique Application ID (format: APP-LM-YYYY-XXXXXX)
- System shall set application status to "Submitted – Awaiting Payment & Scrutiny"

4.3.2 Automatic Routing (LMO vs. GATC)

Requirements:

- System shall automatically route applications based on instrument type
- System shall maintain GATC-approved instrument list (configured by Central Admin per GATC Rules, 2013 Schedule I & II)
- **GATC-Approved Instruments:**
 - Analytical Balances, Microbalances
 - Medical Baby Scales, Hospital Scales
 - Coriolis/Ultrasonic Flow Meters
 - Custody Transfer Meters (oil/gas)
 - Railway Weighbridges
 - Aircraft Fuel Dispensers
 - Energy Meters (electricity)
 - Radiation Meters (X-ray, CT scanners)
 - Speedometers (automotive testing)
 - Breath Alcohol Analyzers
- **Routing Logic:**
 - IF instrument_type IN GATC_APPROVED_LIST:
 - Route to GATC
 - Show list of accredited GATCs near applicant location (sorted by distance)

- Applicant selects preferred GATC
- Application assigned to selected GATC
- ELSE:
 - Route to LMO
 - Auto-assign to jurisdictional LMO (based on installation pincode)
 - Application assigned to LMO
- System shall log routing decision (audit trail)

4.3.3 Fee Calculation (Display Only for MVP)

Requirements:

- System shall calculate fee dynamically based on:
 - Instrument type (per Rule 14 of Legal Metrology (General) Rules, 2011)
 - Capacity (higher capacity = higher fee)
 - Verification type (initial vs. re-verification)
 - Late fee (if re-verification after expiry; 50% extra per quarter delay per Rule 14)
- System shall display fee breakdown:
 - Base verification fee
 - Late fee (if applicable)
 - GATC lab charges (if applicable; separate from LMO fee)
 - Total fee
- System shall display payment instructions:
 - "Please pay ₹[total] at District Legal Metrology Office / GATC counter"
 - "Bring this application printout with payment challan"
- System shall allow applicant to download application PDF (for manual payment)
- System shall NOT integrate payment gateway (MVP; manual payment only)

4.3.4 Payment Receipt Upload

Requirements:

- System shall allow applicants to upload payment receipt after manual payment
- Upload interface shall include:
 - Application ID (auto-filled from "My Applications" list)
 - Payment receipt file upload (PDF/JPG, max 5MB)
 - Payment amount (numeric; must match calculated fee)
 - Payment date (date picker)
 - Payment counter (text: e.g., "District Legal Metrology Office, Mumbai")

- System shall validate:
 - File uploaded (required)
 - Payment amount matches calculated fee ($\pm 5\%$ tolerance for rounding)
- System shall update application status to "Payment Uploaded – Awaiting Scrutiny"
- System shall notify LMO/GATC (via dashboard status change)

4.4 Application Scrutiny & Assignment Module

4.4.1 LMO Scrutiny (Desktop Review)

Requirements:

- System shall display assigned applications in LMO dashboard under "Pending Applications"
- LMO shall be able to view:
 - Applicant details (business name, GSTIN, contact info, address)
 - Instrument details (type, make, model, serial number, capacity, TAC)
 - Uploaded documents (purchase invoice, previous certificate, repair bill)
 - Payment status (receipt uploaded, amount verified)
- LMO shall be able to take actions:
 - **Accept & Schedule:** Application complete & eligible
 - Status changes to "Assigned – Inspection Scheduled"
 - LMO manually informs applicant of appointment (phone/in-person)
 - **Request Clarification:** Documents incomplete/unclear
 - Status changes to "Clarification Asked"
 - LMO adds remark (e.g., "Upload clearer TAC copy")
 - Applicant sees status change in dashboard
 - Applicant must re-upload documents
 - Application paused until resubmission
 - **Reject:** Ineligible (instrument not under Legal Metrology, fake documents)
 - Status changes to "Rejected"
 - LMO selects reason from dropdown (or free text)
 - Applicant notified via dashboard
 - Applicant can appeal to Assistant Controller
- System shall log scrutiny decision (audit trail)

4.4.2 Payment Verification (LMO/GATC)

Requirements:

- System shall allow LMO/GATC to verify uploaded payment receipts
- LMO/GATC shall view:
 - Uploaded payment receipt (PDF/JPG)
 - Calculated fee (from application)
 - Payment amount entered by applicant
- LMO/GATC shall mark payment as:
 - **Verified:** Amount matches, receipt valid
 - Status changes to "Payment Verified – Awaiting Scrutiny"
 - **Rejected:** Amount mismatch, fake receipt
 - Status changes to "Payment Rejected"
 - LMO/GATC adds remark
 - Applicant must re-upload correct receipt
- System shall log verification decision (audit trail)

4.4.3 Inspection Scheduling

Requirements:

- System shall provide calendar view for LMOs/GATCs to schedule inspections
- Calendar shall show:
 - Existing appointments (color-coded by status)
 - Available slots (based on working hours: 9 AM - 6 PM, excluding weekends/holidays)
 - Overbooking prevention (max appointments per slot configurable)
- LMO/GATC shall select slot → confirm appointment
- System shall update application status to "Inspection Scheduled"
- System shall display appointment details in applicant dashboard:
 - Inspector/GATC name
 - Scheduled date & time
 - Contact number (manual communication for MVP)
- System shall allow rescheduling (by LMO/GATC or applicant with approval)

4.5 Inspection & Testing Module

4.5.1 LMO Field Inspection

Requirements:

- System shall provide inspection checklist per instrument type (aligned with OIML R76)
- Checklist shall include:
 - **Physical Inspection:**
 - Manufacturer's seal intact (Yes/No)
 - Display readability (Yes/No)
 - Leveling (for platform scales; Yes/No)
 - Power supply stability (for electronic instruments; Yes/No)
 - Overall condition (dropdown: Good/Fair/Poor)
 - Remarks (free text)
 - **Metrological Tests:**
 - Zero Error Test (observed value, pass/fail)
 - Span Test at 25%, 50%, 75%, 100% capacity (observed values, errors, pass/fail)
 - Eccentricity Test (center, front-left, front-right, rear-left, rear-right; observed values, pass/fail)
 - Discrimination Test (observed value, pass/fail)
 - Repeatability Test (3 readings, variation, pass/fail)
 - **MPE Calculation:**
 - System shall auto-calculate MPE limits based on instrument class (Class I, II, III, IIII per OIML R76)
 - System shall compare observed errors vs. MPE
 - System shall determine overall result: PASS (all within MPE) / FAIL (any exceeds MPE)
- LMO shall be able to:
 - Upload photos (instrument display, seal location, test setup, test weights; JPG/PNG, max 5MB each)
 - Enter seal number (if passed; text, required)
 - Enter GPS coordinates (optional; auto-captured if mobile app used)
 - Enter remarks (free text)
 - Submit inspection report
- System shall update application status to "Inspection Completed – Awaiting Certificate Issuance" (if passed) or "Rejected – Rectification Required" (if failed)
- System shall generate rejection notice (PDF) if failed:

- Specific test failures
- Corrective actions suggested
- Option to reapply after repair

4.5.2 GATC Laboratory Testing

Requirements:

- System shall provide test report template per instrument type (per GATC Rules, 2013)
- Test report shall include:
 - **Instrument Details:** Make, model, serial number, capacity, applicant details
 - **Test Conditions:** Temperature, humidity, vibration levels during testing
 - **Test Results:** Numerical data for all tests (repeatability, linearity, etc.)
 - **Error Analysis:** Observed errors vs. MPE limits
 - **Uncertainty Budget:** Combined uncertainty calculation ($k=2$, 95% confidence)
 - **Conclusion:** "Conforms to Legal Metrology requirements" OR "Does not conform due to [reasons]"
 - **Recommendation:** "Certify" OR "Reject"
 - **Attachments:** Raw data files (CSV/Excel), calibration certificates (PDF), photos (JPG)
- GATC shall be able to:
 - Log instrument receipt (date, time, condition, photos)
 - Upload test report (PDF, required)
 - Upload raw data files (CSV/Excel, optional)
 - Upload calibration certificates (PDF, required)
 - Upload photos (JPG/PNG, optional)
 - Submit test report to LMO
- System shall update application status to "GATC Testing Completed – Awaiting LMO Approval"
- System shall notify LMO (via dashboard status change)

4.5.3 LMO Review of GATC Report

Requirements:

- System shall allow LMO to review GATC test report (PDF)
- LMO shall verify:
 - GATC accreditation valid for this instrument type
 - Test procedures align with OIML/Legal Metrology rules
 - Errors within MPE limits

- Uncertainty values acceptable
- LMO shall be able to take actions:
 - **Approve:** Triggers certificate generation
 - Status changes to "GATC Testing Passed – Awaiting Certificate Issuance"
 - **Request Clarification:** Asks GATC for additional data/retest
 - Status changes to "Clarification Asked – GATC"
 - LMO adds remark
 - GATC must respond with additional data/retest report
 - **Reject:** If report non-compliant
 - Status changes to "Rejected – GATC Report Non-Compliant"
 - LMO selects reason
 - Applicant notified via dashboard
- System shall log approval decision (audit trail)

4.6 Certificate Generation Module

4.6.1 Certificate Drafting (Schedule VIII Format)

Requirements:

- System shall auto-generate certificate in **Schedule VIII format** (per Rule 11 of Legal Metrology (General) Rules, 2011)
- Certificate shall include:
 - Header: "GOVERNMENT OF [STATE NAME] - DEPARTMENT OF LEGAL METROLOGY"
 - Title: "CERTIFICATE OF VERIFICATION (Under Section 24, Legal Metrology Act, 2009)"
 - Certificate Number (format: LM/VER/YYYY/XXXXX)
 - Date of Issue (DD/MM/YYYY)
 - Valid Until (DD/MM/YYYY; auto-calculated per Rule 13 validity periods)
 - Instrument Details:
 - Type (e.g., Electronic Weighing Scale)
 - Make/Model
 - Serial Number
 - Capacity / Division
 - TAC Number (if applicable)
 - Owner Details:
 - Name
 - Business Name

- Address
- GSTIN (if any)
- Mobile Number
- Verification Details:
 - Verified By (LMO name / GATC name)
 - Designation (Legal Metrology Officer / GATC In-charge)
 - Office/Lab (District Legal Metrology Office, [District])
 - Date of Verification (DD/MM/YYYY)
 - Seal Number
 - Stamp Quarter (e.g., Q3-2026 for July-Sept)
 - Result: PASSED (within MPE)
 - Test Report Reference (GATC Report ID, if applicable)
- Remarks (free text from LMO)
- Digital Signature placeholder (text: "Digitally Signed by [LMO Name]" for MVP; DSC later)
- Official Seal/Stamp Image (placeholder for MVP)
- QR Code (encoded with certificate number + basic details)
- Footer Note: "This certificate is valid only until the date mentioned above. Re-verification is mandatory thereafter. Penalty for non-compliance: Up to ₹10,000 (Section 30). Verify authenticity: <https://verify.emaap.gov.in>"
- System shall generate certificate as PDF (A4 size, printable)

4.6.2 Certificate Approval & Issuance

Requirements:

- System shall allow LMO to review draft certificate
- LMO shall verify all details (instrument, owner, test results, seal number)
- LMO shall click "Approve & Issue Certificate" button
- System shall:
 - Generate final PDF certificate
 - Generate QR code (encoded with certificate number + validity + instrument type + owner name)
 - Update application status to "Certificate Issued – Active"
 - Update instrument passport (add new certificate, validity period, seal number)
 - Make certificate available for download in owner dashboard
- System shall NOT integrate digital signatures (MVP; text-only approval)

4.6.3 Certificate Download & Printing

Requirements:

- System shall allow owners to download certificates (PDF)
- Download shall be available from:
 - Owner dashboard ("My Certificates" section)
 - Application tracking page (once status is "Certificate Issued")
- System shall allow unlimited downloads (no restrictions)
- System shall allow printing (PDF optimized for A4 printing)
- System shall include QR code on certificate (scannable by public)
- System shall NOT allow modifications to issued certificates (immutable)

4.7 Public Verification Module

4.7.1 Certificate Search

Requirements:

- System shall provide public portal (no login required)
- Search options:
 - **By Certificate Number:** Enter certificate number → view details
 - **By Instrument Serial Number:** Enter serial number → view all certificates for that instrument
 - **By Owner Name/Business Name:** Enter name → view all instruments registered to that business
- Search results shall display:
 - Certificate Number
 - Issue Date
 - Valid Until
 - Status (Valid/Expired/Revoked; color-coded: Green/Yellow/Red)
 - Instrument Type, Make, Model, Serial Number
 - Owner Name, Business Name, Address
 - Issuing Authority (LMO name, office)
 - Seal Number
 - Test Results Summary (Pass/Fail)
- System shall NOT display:
 - Full instrument passport (summary only)

- Contact details (mobile, email)
- Payment details
- Inspection photos

4.7.2 QR Code Scanning

Requirements:

- System shall provide QR code scanner (mobile-optimized web page)
- User scans QR code on certificate/instrument sticker
- QR code redirects to: `verify.emaap.gov.in/cert/{certificate_number}`
- Verification page displays same details as certificate search (Section 4.7.1)
- System shall validate QR code authenticity (check if certificate number exists in database)
- System shall display warning if QR code invalid/tampered

4.7.3 Discrepancy Reporting

Requirements:

- System shall provide "Report Discrepancy" button on verification page
- Report form shall include:
 - Discrepancy Type (dropdown: Certificate appears fake, Instrument not matching certificate, Seal damaged/missing, Expired certificate in use, Other)
 - Description (free text)
 - Photo upload (JPG/PNG, optional, max 5MB)
 - Reporter Contact (mobile/email, optional)
- System shall create complaint ticket
- System shall assign ticket to jurisdictional LMO (based on instrument location)
- System shall display ticket number to reporter (for tracking)
- System shall allow reporter to track status (via ticket number)
- System shall notify LMO (via dashboard; no SMS/email for MVP)

4.8 Renewal & Re-verification Module

4.8.1 Expiry Tracking (Manual for MVP)

Requirements:

- System shall display certificate validity in owner dashboard
- Dashboard shall show list of instruments with:
 - Instrument details (type, make, model, serial number)

- Certificate number
- Issue Date
- Valid Until
- Status (color-coded: ☐ Valid (>90 days), ☐ Expiring Soon (≤90 days), ☐ Expired)
- System shall NOT send automated reminders (MVP; manual tracking by owner)
- System shall allow filtering by status (Valid/Expiring/Expired)

4.8.2 Renewal Application

Requirements:

- System shall allow owners to apply for re-verification (renewal)
- Renewal application form shall auto-fill:
 - Instrument details (from previous certificate)
 - Last certificate number
 - Previous validity period
 - Previous LMO/GATC details
- Owner shall select:
 - Preferred date (within quarter system: Jan-Mar, Apr-Jun, etc.)
 - Verification mode (LMO field visit or GATC; same as previous or change)
- System shall calculate fee:
 - Base fee (same as initial verification)
 - Late fee (if expired; auto-calculated per Rule 14: 50% per quarter delay)
- System shall display fee breakdown + payment instructions
- Owner shall download application PDF → pay manually → upload receipt
- Inspection & testing process same as initial verification (Section 4.5)
- New certificate issued:
 - Validity extended (12 months from expiry date, not from re-verification date)
 - Previous certificate archived (accessible in instrument passport history)
 - New seal number logged

4.8.3 Grace Period & Late Fee (Rule 14)

Requirements:

- System shall implement grace period rules:
 - Instruments verified in a quarter (e.g., Jan-Mar) can be re-verified any time within that quarter without penalty
 - Example: Certificate valid until 31-Mar-2027; re-verified on 15-Mar-2027 → no late fee

- System shall calculate late fee per Rule 14:
 - First quarter delay (e.g., Apr-Jun for Mar expiry): 50% extra fee
 - Second quarter delay (e.g., Jul-Sep): 100% extra fee (50% × 2)
 - Third quarter delay (e.g., Oct-Dec): 150% extra fee (50% × 3)
 - Beyond 1 year: Treated as new verification (full fee + possible penalty)
- System shall display late fee breakdown in renewal application

4.9 Grievance & Appeals Module

4.9.1 Grievance Submission (Owner)

Requirements:

- System shall allow owners to submit grievances
- Grievance form shall include:
 - Application ID (dropdown from "My Applications")
 - Grievance Type (dropdown: LMO did not visit, GATC damaged instrument, Certificate error, Payment dispute, Other)
 - Description (free text)
 - Supporting documents upload (PDF/JPG, optional)
- System shall create grievance ticket
- System shall assign ticket to:
 - Jurisdictional LMO (for LMO-related issues)
 - GATC in-charge (for GATC-related issues)
 - Assistant Controller (for payment disputes, certificate errors)
- System shall display ticket number to owner (for tracking)

4.9.2 Grievance Resolution (LMO/GATC/AC)

Requirements:

- System shall allow LMO/GATC/AC to view assigned grievances
- User shall be able to:
 - View grievance details (type, description, documents)
 - Add remarks/resolution notes
 - Upload resolution documents (PDF, optional)
 - Mark grievance as "Resolved"
- System shall update grievance status to "Resolved"
- System shall notify owner (via dashboard status change)

- System shall allow owner to escalate if not resolved within 7 days (to Assistant Controller)

4.9.3 Appeals (Rejected Applications/Penalties)

Requirements:

- System shall allow owners to appeal rejected applications or penalties
- Appeal form shall include:
 - Application ID / Penalty ID
 - Reason for appeal (free text)
 - Supporting documents upload (PDF/JPG, required)
- System shall assign appeal to Assistant Controller (district-level)
- Assistant Controller shall:
 - Review appeal (application details, rejection reason, owner's argument)
 - Upload hearing notes (PDF, optional)
 - Make decision: Uphold rejection / Overturn rejection / Modify penalty
 - Upload order (PDF, required)
- System shall update application/penalty status based on decision
- System shall notify owner (via dashboard)

4.10 Seizure Management Module (Section 15)

4.10.1 Seizure Proposal (LMO)

Requirements:

- System shall allow LMOs to propose instrument seizures (under Section 15 of Legal Metrology Act, 2009)
- Seizure form shall include:
 - Instrument details (type, make, model, serial number, owner)
 - Reason for seizure (dropdown: Fake certificate, Expired certificate, Tampered seal, Accuracy failure, Other)
 - Seizure location (address, GPS coordinates if available)
 - Seizure date & time
 - Photos upload (JPG/PNG, required: instrument, certificate, seal)
 - Witness details (name, contact, optional)
- System shall generate seizure memo (PDF)
- System shall assign seizure proposal to Assistant Controller (for approval)

4.10.2 Seizure Approval (Assistant Controller)

Requirements:

- System shall allow Assistant Controller to review seizure proposals
- AC shall view:
 - Seizure details (instrument, owner, reason, photos)
 - LMO remarks
- AC shall be able to:
 - Approve seizure
 - Reject seizure (with reason)
 - Request additional information
- System shall update seizure status:
 - "Approved" → Instrument seized (LMO takes custody)
 - "Rejected" → Instrument returned to owner
 - "Additional Information Asked" → LMO must provide more details
- System shall notify LMO (via dashboard)

4.10.3 Seizure Disposal

Requirements:

- System shall allow LMOs to update seizure disposal status
- Disposal options:
 - Returned to owner (after rectification)
 - Auctioned (as per government rules)
 - Destroyed (if beyond repair)
- System shall log disposal details (date, method, amount if auctioned)
- System shall update instrument status to "Seized – Disposed"

4.11 Reporting & MIS Module

4.11.1 Dashboard Reports (All Roles)

Requirements:

- System shall provide basic reports in each dashboard:
 - Owner Dashboard:
 - My Instruments (count by status: Valid, Expiring, Expired)

- My Applications (count by status: Submitted, Under Inspection, Certified, Rejected)
- My Certificates (count, list with validity)
- LMO Dashboard:
 - Pending Applications (count, list)
 - Inspections Completed (count this month, pass/fail ratio)
 - Certificates Issued (count this month)
 - Pendency Report (applications pending >7 days, >15 days)
- GATC Dashboard:
 - Tests Pending (count, list)
 - Tests Completed (count this month, pass/fail ratio)
 - Average Testing Time (days)
- Assistant Controller Dashboard:
 - District Overview (total instruments, verified count, compliance %)
 - LMO Performance (inspections per LMO, pendency)
 - Grievances (pending, resolved, average resolution time)
 - Seizures (proposed, approved, rejected)
- Central Admin Dashboard:
 - National Overview (total instruments, verified count by state)
 - GATC Registry (count, accreditation validity)
 - Public Verification Stats (QR scans, discrepancy reports)

4.11.2 Export Functionality

Requirements:

- System shall allow export of reports to PDF/Excel
- Export options:
 - Filter by date range
 - Filter by instrument type
 - Filter by status
- System shall generate downloadable file (PDF for reports, Excel for data tables)

5. End-to-End Application Flow

5.1 Flow Diagram (Text Representation)

PHASE 1: OWNER REGISTRATION

- 1.1 Owner visits portal → clicks "Register"
- 1.2 Fills registration form (business details, documents)
- 1.3 System validates → generates User ID → displays password
- 1.4 Owner logs in → changes password
- 1.5 Owner registers instruments (pre-registration)



PHASE 2: APPLICATION SUBMISSION

- 2.1 Owner clicks "Apply for Verification"
- 2.2 Selects instrument → fills application form
- 2.3 System auto-routes (LMO or GATC based on instrument)
- 2.4 System calculates fee → displays payment instructions
- 2.5 Owner downloads application PDF → pays manually
- 2.6 Owner uploads payment receipt → application submitted



PHASE 3: SCRUTINY & ASSIGNMENT

- 3.1 LMO/GATC views application in dashboard
- 3.2 LMO/GATC verifies payment receipt
- 3.3 LMO scrutinizes application (documents, TAC)
- 3.4 LMO takes action:
 - Accept → schedule inspection
 - Request clarification → owner resubmits documents
 - Reject → owner can appeal
- 3.5 LMO/GATC schedules inspection (calendar)
- 3.6 System notifies owner (dashboard status change)



PHASE 4: INSPECTION & TESTING

- 4.1 LMO conducts field inspection (or GATC lab testing)
- 4.2 LMO/GATC performs tests (per OIML R76 / GATC Rules)
- 4.3 LMO/GATC uploads inspection report + photos
- 4.4 If GATC: submits report to LMO for approval
- 4.5 LMO reviews GATC report → approves/rejects
- 4.6 System updates status: Inspection Completed



PHASE 5: CERTIFICATE GENERATION

- 5.1 System auto-generates Schedule VIII certificate (PDF)
- 5.2 LMO reviews draft certificate
- 5.3 LMO clicks "Approve & Issue"
- 5.4 System generates final PDF + QR code
- 5.5 System updates instrument passport
- 5.6 Certificate available for download in owner dashboard



PHASE 6: PUBLIC VERIFICATION

- 6.1 Public visits verification portal (no login)
- 6.2 Searches by certificate number / serial number / name
OR scans QR code on certificate/instrument
- 6.3 System displays certificate summary (validity, owner)
- 6.4 Public can report discrepancy (if fake/expired)
- 6.5 Complaint assigned to LMO → investigated



PHASE 7: RENEWAL (RE-VERIFICATION)

- 7.1 Owner tracks expiry in dashboard (manual for MVP)
- 7.2 Owner clicks "Apply for Re-verification"
- 7.3 System auto-fills previous certificate details
- 7.4 System calculates fee (base + late fee if expired)
- 7.5 Owner pays manually → uploads receipt
- 7.6 Inspection & testing (same as Phase 4)
- 7.7 New certificate issued (validity extended 12 months)
- 7.8 Previous certificate archived in instrument passport

5.2 State Machine (Application Status Flow)

[Draft]

↓ (owner saves application)

[Submitted]

↓ (owner uploads payment receipt)

[Payment Uploaded]

↓ (LMO/GATC verifies payment)

[Payment Verified]

↓ (LMO scrutinizes application)

[Under Scrutiny]

↳ [Clarification Asked] → [Documents Resubmitted] → [Under Scrutiny]

↳ [Rejected] → [Appeal Filed] → [Appeal Decision] → [Approved or Rejected]

↓ (LMO accepts & schedules)

[Assigned - Inspection Scheduled]

↓ (inspection completed)

[Inspection Completed]

↳ [Passed] → [Certificate Under Issuance] → [Certificate Issued - Active]

↳ [Failed] → [Rejection Notice] → [Rectification Required] → [Reapply]

Renewal Flow:

[Active]

↓ (owner manually tracks expiry)

[Expiring Soon] (☒ in dashboard; ≤90 days)

↓ (on expiry date)

[Expired] (☒ in dashboard)

↓ (owner applies for renewal)

[Renewal Application Submitted]

↓ (payment uploaded & verified)

```
[Payment Verified]
  ↓ (inspection scheduled & completed)
[Inspection Completed]
  ↓ (passed)
[Certificate Re-Issued - Active]
```

6. Dashboard Specifications

6.1 Instrument Owner Dashboard

6.1.1 Dashboard Widgets

- **My Instruments:**
 - Total count
 - Breakdown by status (Valid, Expiring Soon, Expired, Under Inspection, Decommissioned)
 - Quick link to "Register New Instrument"
- **My Applications:**
 - Total count
 - Breakdown by status (Submitted, Under Scrutiny, Inspection Scheduled, Certified, Rejected)
 - Quick link to "Apply for Verification"
- **My Certificates:**
 - Total count
 - List with validity (color-coded)
 - Quick link to "Download Certificates"
- **Expiring Soon:**
 - List of instruments with certificates expiring within 90 days
 - Quick link to "Apply for Re-verification"

6.1.2 Navigation Menu

- Dashboard (Home)
- My Instruments
 - Register New Instrument
 - View All Instruments
 - Decommission Instrument
- My Applications
 - Apply for Verification

- Track Applications
- Upload Payment Receipt
- My Certificates
 - Download Certificates
 - Instrument Passport
- Grievances
 - Submit Grievance
 - Track Grievances
- Account Settings
 - Change Password
 - Update Profile
 - Logout

6.1.3 Key Pages

- **Register Instrument:** Form (Section 4.2.1)
- **Apply for Verification:** Form (Section 4.3.1)
- **Track Application:** Timeline view with status updates
- **Download Certificates:** List with download buttons (PDF)
- **Instrument Passport:** Complete history (all certificates, inspections, seals)
- **Submit Grievance:** Form (Section 4.9.1)

6.2 LMO Dashboard

6.2.1 Dashboard Widgets

- **Pending Applications:**
 - Count (new, under scrutiny, scheduled)
 - List with priority (expiring instruments highlighted)
- **Inspection Schedule:**
 - Calendar view (today's appointments, upcoming)
 - Quick link to "Schedule Inspection"
- **My Performance:**
 - Inspections completed (this month)
 - Certificates issued (this month)
 - Average time to certify (days)
- **Pendency Alert:**

- Applications pending >7 days (count)
- Applications pending >15 days (count)

6.2.2 Navigation Menu

- Dashboard (Home)
- Applications
 - Pending Applications
 - My Applications (all)
 - Scrutinize Application
- Inspections
 - Schedule Inspection
 - Upload Inspection Report
 - Inspection History
- Certificates
 - Issue Certificates
 - Certificate History
- Seizures
 - Propose Seizure
 - Seizure History
- Grievances
 - Assigned Grievances
 - Resolve Grievance
- Reports
 - Pendency Report
 - Performance Report
- Account Settings
 - Change Password
 - Logout

6.2.3 Key Pages

- **Scrutinize Application:** View details + action buttons (Accept/Request Clarification/Reject)
- **Schedule Inspection:** Calendar + slot selection
- **Upload Inspection Report:** Form (Section 4.5.1) with checklist, photo upload, seal number entry
- **Issue Certificate:** Review draft + "Approve & Issue" button

- **Propose Seizure:** Form (Section 4.10.1) with photo upload
- **Resolve Grievance:** View complaint + add resolution notes

6.3 GATC Dashboard

6.3.1 Dashboard Widgets

- **Test Queue:**
 - Pending tests (count)
 - In-progress tests (count)
 - Completed tests (count)
- **Lab Capacity:**
 - Available slots (this week)
 - Booked slots (this week)
- **Performance Metrics:**
 - Average testing time (days)
 - Pass/fail ratio
 - LMO approval rate

6.3.2 Navigation Menu

- Dashboard (Home)
- Testing Requests
 - Pending Requests
 - All Requests
 - Schedule Appointment
- Test Reports
 - Upload Test Report
 - Test Report History
- Lab Management
 - Manage Slots
 - Instrument Receipt Log
- Accreditation
 - View Accreditation Status
 - Renewal Alert
- Reports
 - Performance Report

- Account Settings
 - Change Password
 - Logout

6.3.3 Key Pages

- **Schedule Appointment:** Calendar + slot selection
- **Upload Test Report:** Form (Section 4.5.2) with PDF upload, raw data upload, calibration certificates
- **Instrument Receipt Log:** Log receipt details (date, time, condition, photos)
- **View Accreditation Status:** Accreditation number, categories approved, validity period

6.4 Assistant Controller Dashboard

6.4.1 Dashboard Widgets

- **District Overview:**
 - Total instruments (district)
 - Verified count
 - Compliance % (verified/total)
- **LMO Performance:**
 - Inspections per LMO (table)
 - Average time to certify (per LMO)
 - Pendency (per LMO)
- **Grievances:**
 - Pending grievances (count)
 - Resolved grievances (count)
 - Average resolution time (days)
- **Seizures:**
 - Pending approvals (count)
 - Approved seizures (count)
 - Rejected seizures (count)

6.4.2 Navigation Menu

- Dashboard (Home)
- Applications
 - View All Applications (district)

- Appeals
- LMO Management
 - Create LMO Account
 - View LMO Performance
 - Pendency Report
- Grievances
 - Assigned Grievances
 - Escalated Grievances
 - Resolve Grievance
- Seizures
 - Approve Seizures
 - Seizure History
- Reports
 - District MIS Report
 - Export to PDF/Excel
- Account Settings
 - Change Password
 - Logout

6.4.3 Key Pages

- **Appeals:** View appeal details + hearing notes upload + decision buttons
- **Create LMO Account:** Form (Section 4.1.4) with employee details, jurisdiction assignment
- **Approve Seizures:** Review seizure proposal + approve/reject buttons
- **District MIS Report:** Summary statistics + export button

6.5 Central Administrator Dashboard

6.5.1 Dashboard Widgets

- **National Overview:**
 - Total instruments (all states)
 - Verified count (by state)
 - Compliance % (by state)
- **GATC Registry:**
 - Total GATCs (active)
 - GATCs expiring soon (count)

- **Public Verification:**
 - QR scans (this month)
 - Discrepancy reports (this month)
- **User Management:**
 - Total users (by role)
 - Active users (this month)

6.5.2 Navigation Menu

- Dashboard (Home)
- GATC Management
 - Create GATC Account
 - View GATC Registry
 - Configure GATC Instrument List
- User Management
 - Create Assistant Controller Account
 - View All Users
 - Deactivate/Reactivate Users
- Configuration
 - Master Data (States, Districts, Pincodes)
 - Instrument Types Catalog
 - Validity Periods (Rule 13)
- Policy Documents
 - Upload Circulars/Notifications
 - View Policy Library
- Reports
 - National Overview Report
 - Public Verification Stats
 - Export to PDF/Excel
- Account Settings
 - Change Password
 - Logout

6.5.3 Key Pages

- **Create GATC Account:** Form (Section 4.1.4) with accreditation details, categories approved
- **Configure GATC Instrument List:** Checklist of instruments (per GATC Rules Schedule I & II)
- **Upload Circulars/Notifications:** PDF upload + description
- **National Overview Report:** Heatmap by state + drill-down to district

6.6 Public Verification Portal

6.6.1 Homepage

- **Search Bar** (prominent, center of page):
 - Search by Certificate Number
 - Search by Instrument Serial Number
 - Search by Owner/Business Name
- **QR Code Scanner** button (mobile-optimized)
- **Quick Links:**
 - Report Discrepancy
 - Download Sample Formats
 - FAQs & Help
 - Contact Us

6.6.2 Search Results Page

- Certificate details (Section 4.7.1)
- Status badge (Valid/Expired/Revoked; color-coded)
- "Report Discrepancy" button
- "Print/Save" button (PDF)

6.6.3 Report Discrepancy Page

- Form (Section 4.7.3) with discrepancy type, description, photo upload
- Ticket number display (after submission)
- Track Ticket Status (by ticket number)

6.6.4 Downloads Page

- Schedule VIII Certificate Format (PDF)
- Fee Schedule (PDF)
- User Manuals (PDF)

- Legal Metrology Act Summary (PDF)
- FAQs (PDF)

7. Certificate & Document Specifications

7.1 Certificate of Verification (Schedule VIII Format)

7.1.1 Layout Requirements

- **Size:** A4 (210mm × 297mm)
- **Orientation:** Portrait
- **Margins:** Top 25mm, Bottom 20mm, Left 20mm, Right 20mm
- **Font:** Arial or Times New Roman, 12pt for body, 14pt bold for headers
- **Header:** Government emblem (placeholder for MVP), "GOVERNMENT OF [STATE NAME]", "DEPARTMENT OF LEGAL METROLOGY"
- **Footer:** Disclaimer note, verification URL

7.1.2 Content Requirements (Exact Fields)

GOVERNMENT OF [STATE NAME]
DEPARTMENT OF LEGAL METROLOGY

CERTIFICATE OF VERIFICATION
(Under Section 24, Legal Metrology Act, 2009)

Certificate No: LM/VER/YYYY/XXXXX
Date of Issue: DD/MM/YYYY
Valid Until: DD/MM/YYYY

INSTRUMENT DETAILS
Type: [Instrument Type]
Make/Model: [Make and Model]
Serial No: [Serial Number]
Capacity: [Capacity] / Division: [Division]
TAC No: [Type Approval Certificate Number, if applicable]

OWNER DETAILS
Name: [Owner Name]
Business Name: [Business Name]
Address: [Full Address]
GSTIN: [GSTIN, if applicable]
Mobile: [Mobile Number]

VERIFICATION DETAILS
Verified By: [LMO Name / GATC Name]
Designation: Legal Metrology Officer / GATC In-charge
Office/Lab: District Legal Metrology Office, [District]
Date of Verification: DD/MM/YYYY

Seal No: [Seal Number]

Stamp Quarter: Q[1-4]-YYYY (e.g., Q3-2026 for July-Sept)

Result: PASSED (within Maximum Permissible Error)

Test Report Reference: [GATC Report ID, if applicable]

Remarks: [LMO Remarks, if any]

[Digital Signature Placeholder: "Digitally Signed by [LMO Name]"]

[Official Seal/Stamp Image Placeholder]

[QR Code: Encoded with Certificate Number + Validity + Instrument Type + Owner Name]

NOTE: This certificate is valid only until the date mentioned above. Re-verification

Penalty for non-compliance: Up to ₹10,000 (Section 30, Legal Metrology Act, 2009)

Verify authenticity: <https://verify.emaap.gov.in>

7.1.3 QR Code Requirements

- **Size:** 2cm × 2cm (minimum scannable size)
- **Error Correction:** Medium (15% error tolerance)
- **Encoded Data:**
 - Certificate Number
 - Valid Until date
 - Instrument Type
 - Owner Name
 - Verification URL (https://verify.emaap.gov.in/cert/{certificate_number})
- **Placement:** Bottom-right corner of certificate (above footer note)

7.1.4 File Format

- **Format:** PDF (A4, printable)
- **File Size:** Maximum 500KB (optimized for download)
- **Security:** Read-only (no editing allowed; password protection optional for future)
- **Download:** Unlimited downloads allowed from owner dashboard

7.2 Inspection Report (LMO)

7.2.1 Layout Requirements

- **Size:** A4 (210mm × 297mm)
- **Orientation:** Portrait
- **Sections:**
 - Instrument Details (same as certificate)

- Physical Inspection Checklist (Yes/No/Remarks)
- Metrological Test Results (table with observed values, errors, MPE limits, pass/fail)
- Overall Result (PASS/FAIL)
- Seal Number (if passed)
- Inspector Details (name, badge number, signature)
- Photos (embedded or attached as annexure)

7.2.2 File Format

- **Format:** PDF (generated from system; LMO cannot modify after submission)
- **Photos:** Embedded in PDF or separate JPG files (max 5MB each)

7.3 Test Report (GATC)

7.3.1 Layout Requirements

- **Size:** A4 (210mm × 297mm)
- **Orientation:** Portrait
- **Sections:**
 - Instrument Details (same as certificate)
 - Test Conditions (temperature, humidity, vibration)
 - Test Results (detailed numerical data, tables, graphs if applicable)
 - Error Analysis (observed errors vs. MPE)
 - Uncertainty Budget (combined uncertainty, k=2, 95% confidence)
 - Conclusion (Conforms/Does not conform)
 - Recommendation (Certify/Reject)
 - GATC Details (name, accreditation number, authorized signatory, signature)
 - Attachments (raw data files, calibration certificates, photos)

7.3.2 File Format

- **Format:** PDF (uploaded by GATC; system does not generate)
- **Attachments:** CSV/Excel (raw data), PDF (calibration certificates), JPG (photos)

7.4 Payment Receipt (Manual for MVP)

7.4.1 Layout Requirements (Government Counter Receipt)

- **Size:** A5 (148mm × 210mm) or thermal printer roll (as per government office standard)
- **Fields:**
 - Receipt Number (government office serial number)
 - Application ID
 - Applicant Name
 - Instrument Type
 - Fee Amount (₹)
 - Payment Date
 - Payment Counter (office name)
 - Cashier/Officer Name & Signature
 - Government Office Seal

7.4.2 Upload Requirements

- **Format:** PDF or JPG (scanned copy)
- **File Size:** Maximum 5MB
- **Quality:** Readable text (minimum 150 DPI scan)

7.5 Rejection Notice

7.5.1 Layout Requirements

- **Size:** A4 (210mm × 297mm)
- **Sections:**
 - Application ID
 - Instrument Details
 - Owner Details
 - Reason for Rejection (dropdown or free text from LMO)
 - Specific Test Failures (if failed inspection; table with test name, observed value, MPE limit, error)
 - Corrective Actions Suggested (free text)
 - Option to Reapply (instructions)
 - LMO Details (name, office, contact)
 - Date

7.5.2 File Format

- **Format:** PDF (generated by system)
- **Download:** Available in owner dashboard

8. Business Rules & Validation Logic

8.1 Instrument Type Validation

8.1.1 GATC-Approved Instruments (Mandatory Routing)

Rule: If instrument type is in GATC-approved list (per GATC Rules, 2013 Schedule I & II), application **MUST** route to GATC.

GATC-Approved List:

- Analytical Balances
- Microbalances
- Medical Baby Scales
- Hospital Scales
- Coriolis Flow Meters
- Ultrasonic Flow Meters
- Custody Transfer Meters (oil/gas)
- Railway Weighbridges
- Aircraft Fuel Dispensers
- Energy Meters (electricity)
- Radiation Meters (X-ray, CT scanners)
- Speedometers (automotive testing)
- Breath Alcohol Analyzers

Validation:

- System shall check instrument type against GATC-approved list during application submission
- If match → force routing to GATC (applicant must select GATC from list)
- If no match → route to LMO (auto-assign by pincode)

8.1.2 Type Approval Certificate (TAC) Requirement

Rule: New instruments MUST have TAC uploaded (per Legal Metrology (General) Rules, 2011).

Validation:

- If verification type = "Initial Verification" → TAC upload required
- If verification type = "Re-verification" → TAC optional (previous certificate sufficient)
- If TAC not uploaded for initial verification → application cannot be submitted (validation error)

8.2 Fee Calculation Rules

8.2.1 Base Fee (Per Rule 14)

Rule: Fee varies by instrument type and capacity.

Sample Fee Structure (configurable per state):

Instrument Type	Capacity Range	Base Fee (₹)
Electronic Weighing Scale	Up to 15 kg	50
Electronic Weighing Scale	15-100 kg	100
Electronic Weighing Scale	Above 100 kg	200
Platform Scale	Up to 100 kg	100
Platform Scale	100-500 kg	200
Platform Scale	Above 500 kg	500
Weighbridge	Up to 50 tonnes	300
Weighbridge	50-100 tonnes	500
Weighbridge	Above 100 tonnes	1000
Fuel Dispenser	Per nozzle	100
Tank Lorry	Up to 10,000 L	200
Tank Lorry	Above 10,000 L	500
Analytical Balance	Any	2000 (GATC fee) + 200 (LMO approval)
Flow Meter	Any	3000 (GATC fee) + 300 (LMO approval)

Validation:

- System shall calculate base fee based on instrument type + capacity
- System shall display fee breakdown before application submission

8.2.2 Late Fee (Per Rule 14)

Rule: Re-verification after expiry attracts 50% extra fee per quarter delay.

Calculation:

- Determine expiry date from previous certificate
- Calculate quarters delayed:
 - Quarter 1: Expiry date + 1 day to expiry date + 3 months
 - Quarter 2: Expiry date + 3 months + 1 day to expiry date + 6 months
 - Quarter 3: Expiry date + 6 months + 1 day to expiry date + 9 months
 - Quarter 4: Expiry date + 9 months + 1 day onwards
- Late fee = Base fee $\times$ 50% $\times$ number of quarters delayed
- Maximum late fee: 200% of base fee (beyond 1 year, treat as new verification)

Example:

- Certificate expired on 31-Mar-2026
- Re-verification applied on 15-Aug-2026
- Quarters delayed: Q1 (Apr-Jun) + Q2 (Jul-Sep) = 2 quarters
- Base fee: ₹100
- Late fee: ₹100 $\times$ 50% $\times$ 2 = ₹100
- Total fee: ₹100 (base) + ₹100 (late) = ₹200

Validation:

- System shall auto-calculate late fee if re-verification application submitted after expiry
- System shall display late fee breakdown separately

8.3 Validity Period Rules (Per Rule 13)

8.3.1 Standard Validity Periods

Rule: Certificates valid for fixed periods based on instrument type.

Standard Validity Table:

Instrument Type	Validity Period
Electronic Weighing Scales	12 months
Platform Scales	12 months
Weighbridges (non-railway)	12 months
Fuel Dispensers (petrol/diesel)	12 months

Instrument Type	Validity Period
Tank Lorries	24 months
Counter Machines	24 months
Auto/Taxi Fare Meters	12 months
Water Meters	24 months
Gas Meters (domestic)	24 months
Analytical Balances	12 months
Flow Meters	12 months
Energy Meters	24 months

Validation:

- System shall auto-calculate "Valid Until" date based on instrument type
- Valid Until = Verification Date + Validity Period (months)
- Example: Verification on 15-Jan-2026, validity 12 months → Valid Until 14-Jan-2027

8.3.2 Quarter System for Re-verification

Rule: Instruments verified in a quarter can be re-verified any time within that quarter without late fee.

Quarters:

- Q1: January - March
- Q2: April - June
- Q3: July - September
- Q4: October - December

Example:

- Certificate valid until 31-Mar-2027 (Q1)
- Re-verification on 15-Mar-2027 → No late fee (within same quarter)
- Re-verification on 15-Apr-2027 → Late fee applies (Q2; 1 quarter delay)

Validation:

- System shall check if re-verification application submitted within same quarter as expiry
- If yes → no late fee
- If no → calculate late fee per Section 8.2.2

8.4 MPE (Maximum Permissible Error) Rules

8.4.1 MPE Limits (Per OIML R76)

Rule: Weighing instruments must operate within MPE limits based on instrument class and load.

Instrument Classes:

- **Class I:** Special accuracy (analytical balances, precision balances)
- **Class II:** High accuracy (laboratory balances, jewelry scales)
- **Class III:** Medium accuracy (commercial scales, platform scales)
- **Class IIII:** Ordinary accuracy (weighbridges, industrial scales)

MPE Table (Class III Example):

Load Range	MPE ($\pm$)
0 to 500 divisions	0.5 divisions
500 to 2000 divisions	1.0 divisions
Above 2000 divisions	1.5 divisions

Example:

- **Instrument:** Electronic scale, capacity 100 kg, division 10 g
- **Total divisions:** 100 kg / 10 g = 10,000 divisions
- **Load:** 50 kg (5,000 divisions)
- **MPE:** 1.5 divisions = $1.5 \times 10 \text{ g} = 15 \text{ g}$
- **Observed error must be $\leq 15 \text{ g}$ for PASS**

Validation:

- System shall auto-calculate MPE based on instrument class, capacity, division, and test load
- System shall compare observed error vs. MPE
- If observed error $\leq$ MPE $\rightarrow$ PASS
- If observed error $>$ MPE $\rightarrow$ FAIL

8.4.2 Test Point Requirements

Rule: Minimum test points required for span test.

Requirements:

- **Span Test:** Minimum 4 test points (25%, 50%, 75%, 100% of capacity)
- **Eccentricity Test:** Minimum 5 positions (center, front-left, front-right, rear-left, rear-right)

- **Repeatability Test:** Minimum 3 consecutive readings at same load

Validation:

- System shall require LMO/GATC to enter test results for all mandatory test points
- If any test point missing → inspection report cannot be submitted (validation error)

8.5 Application Status Transition Rules

8.5.1 Valid Status Transitions

Rule: Application status can only transition through predefined paths.

Valid Transitions:

- Draft → Submitted
- Submitted → Payment Uploaded
- Payment Uploaded → Payment Verified
- Payment Verified → Under Scrutiny
- Under Scrutiny → Assigned / Clarification Asked / Rejected
- Clarification Asked → Documents Resubmitted → Under Scrutiny
- Assigned → Inspection Scheduled
- Inspection Scheduled → Inspection Completed
- Inspection Completed → Certificate Under Issuance / Rejected
- Certificate Under Issuance → Certificate Issued
- Rejected → Appeal Filed → Appeal Decision → Approved / Rejected

Invalid Transitions (system shall prevent):

- Submitted → Certificate Issued (skip scrutiny/inspection)
- Inspection Completed → Certificate Issued (skip LMO approval)
- Rejected → Certificate Issued (must go through appeal)

Validation:

- System shall enforce status transition rules
- If invalid transition attempted → display error message

8.6 User Access Control Rules

8.6.1 Data Isolation by Jurisdiction

Rule: LMOs can only access applications in their assigned jurisdiction (pincode range).

Validation:

- System shall filter applications by LMO's assigned pincode range
- If LMO attempts to access application outside jurisdiction → display "Access Denied" error

8.6.2 Certificate Immutability

Rule: Issued certificates cannot be modified (immutable).

Validation:

- System shall not allow any edits to certificate after status = "Certificate Issued"
- If correction needed (e.g., typo in owner name):
 - LMO must issue new certificate (with new certificate number)
 - Previous certificate marked as "Superseded" in instrument passport
 - Reason for re-issuance logged (audit trail)

9. Security & Compliance Requirements

9.1 Authentication & Authorization

9.1.1 Password Policy

Requirements:

- Minimum 8 characters
- At least 1 uppercase letter (A-Z)
- At least 1 lowercase letter (a-z)
- At least 1 number (0-9)
- At least 1 special character (!@#\$%^&*()_+-=[]{}|;:,.<>?)
- Cannot reuse last 5 passwords
- Must change every 90 days (system prompts on login)
- Account locked after 5 consecutive failed login attempts (unlock after 30 minutes or admin reset)

9.1.2 Session Management

Requirements:

- Session timeout after 30 minutes of inactivity
- Session invalidated on logout
- Session invalidated on password change
- Concurrent sessions allowed (same user can login from multiple devices)
- Session token stored securely (encrypted, HTTP-only cookie)

9.1.3 Role-Based Access Control (RBAC)

Requirements:

- System shall enforce RBAC at API level (not just UI)
- Each API endpoint shall check user role before processing request
- Example: Only LMOs can access `/api/inspection/submit` endpoint
- Example: Only Central Admin can access `/api/gatc/configure-instrument-list` endpoint
- Unauthorized access → HTTP 403 Forbidden response

9.2 Data Security

9.2.1 Data Encryption (Future Requirement; Not for MVP)

Note: Not required for MVP, but document for future integration.

Requirements:

- All data in transit encrypted via HTTPS (TLS 1.2 or higher)
- All sensitive data at rest encrypted (AES-256):
 - Passwords (hashed with bcrypt/argon2)
 - Personal information (Aadhaar, GSTIN, PAN)
 - Certificate PDFs (optional)
- Encryption keys managed via secure key management service (KMS)

9.2.2 Audit Logging

Requirements:

- System shall log all critical actions:
 - User login/logout (timestamp, IP address, user ID)
 - Application status changes (old status, new status, timestamp, user ID)
 - Certificate issuance (certificate number, timestamp, LMO ID)

- Inspection report submission (application ID, timestamp, LMO/GATC ID)
- User management actions (create/update/deactivate user; admin ID, timestamp)
- Seizure proposals/approvals (instrument ID, timestamp, LMO/AC ID)
- Grievance submissions/resolutions (ticket ID, timestamp, user ID)
- Audit logs stored in separate database table (immutable; no edits allowed)
- Audit logs retained for minimum 10 years
- Audit logs exportable to PDF/Excel (for audits)

9.3 Compliance Requirements

9.3.1 Legal Metrology Act Compliance

Requirements:

- System shall comply with Legal Metrology Act, 2009:
 - Section 24: Verification and stamping (core workflow)
 - Section 15: Inspection and seizure (seizure module)
 - Section 27: Compounding of offences (penalty tracking)
 - Section 30: Penalties (displayed on certificate)
- System shall display relevant Act sections in:
 - Certificate footer (Section 30 penalty warning)
 - Help/FAQ section (link to full Act PDF)
 - Seizure module (Section 15 reference)

9.3.2 Legal Metrology (General) Rules, 2011 Compliance

Requirements:

- System shall comply with Legal Metrology (General) Rules, 2011:
 - Rule 11: Certificate format (Schedule VIII)
 - Rule 12: Stamping (seal number logging)
 - Rule 13: Validity periods (auto-calculation)
 - Rule 14: Fees (fee calculation logic)
 - Rule 15: GATC verification (routing logic)
- System shall reference Rules in:
 - Certificate (Rule 11 reference in header)
 - Fee calculation display (Rule 14 reference)
 - Validity period display (Rule 13 reference)

9.3.3 GATC Rules, 2013 Compliance

Requirements:

- System shall comply with Legal Metrology (GATC) Rules, 2013:
 - Schedule I & II: GATC-approved instrument list (routing logic)
 - Rule 6: GATC functions (test report requirements)
 - Rule 8: GATC fees (fee calculation for GATC instruments)
- System shall reference GATC Rules in:
 - GATC dashboard (Schedule I/II reference)
 - Test report template (Rule 6 reference)

9.4 Data Retention & Privacy

9.4.1 Data Retention Period

Requirements:

- System shall retain data for minimum periods:
 - Certificates: 10 years (from issue date)
 - Inspection reports: 10 years (from inspection date)
 - Applications: 10 years (from submission date)
 - Audit logs: 10 years (from log date)
 - User accounts: Indefinite (until deactivated by admin)
 - Grievances: 5 years (from resolution date)
 - Seizure records: 10 years (from disposal date)
- After retention period → data archived (not deleted; moved to cold storage)

9.4.2 Data Privacy (Basic for MVP)

Requirements:

- System shall collect minimum necessary data:
 - Owner registration: Business details, contact info, documents (as per Legal Metrology requirements)
 - Instrument registration: Instrument details, TAC (as per Legal Metrology requirements)
 - No unnecessary data collection (e.g., religion, caste, political affiliation)
- System shall display privacy notice on registration page:
 - "Your data is collected as per Legal Metrology Act, 2009 and will be used only for verification purposes."

- System shall not share data with third parties (except government authorities as per law)
- System shall allow users to view their own data (dashboard exports)
- System shall allow users to request data correction (via grievance module)

10. Implementation Plan & Phases

10.1 Phase 1: MVP (Minimum Viable Product) - Weeks 1-8

10.1.1 Scope

Goal: Launch core functionality with manual processes (no payment gateway, no SMS/email, no digital signatures).

Features:

- User registration & login (all 6 roles)
- Instrument pre-registration
- Application submission (LMO/GATC routing)
- Manual payment receipt upload
- LMO scrutiny & inspection scheduling
- LMO field inspection (web form; mobile app optional)
- GATC laboratory testing (test report upload)
- Certificate generation (Schedule VIII PDF; text-only approval)
- Public verification portal (search + QR scan)
- Basic dashboards (all roles)
- Audit logging (basic)

Out of Scope:

- Online payment gateway
- SMS/Email notifications
- Digital signatures (DSC/eSign)
- Mobile app (use responsive web for MVP)
- Advanced analytics/MIS
- API integrations (UIDAI, BharatKosh, NIC SMS)

10.1.2 Deliverables

- Fully functional web application (responsive design)
- Database schema (all tables: users, instruments, applications, certificates, inspections, etc.)
- API documentation (for future mobile app integration)
- User manuals (PDF for each role)
- Admin configuration guide (GATC instrument list, validity periods, fee structure)
- Test cases & test results (unit testing, integration testing)
- Deployment guide (server setup, database setup, environment variables)

10.1.3 Timeline

Week	Tasks
Week 1	Requirement finalization, database schema design, UI/UX wireframes
Week 2-3	Backend development (user management, instrument registration, application submission)
Week 4-5	Backend development (scrutiny, inspection, certificate generation)
Week 6	Frontend development (all dashboards, public portal)
Week 7	Integration testing, bug fixes, security testing
Week 8	UAT (user acceptance testing), documentation, deployment to staging

10.2 Phase 2: Enhanced Features - Weeks 9-16

10.2.1 Scope

Goal: Add automation and integrations to reduce manual effort.

Features:

- Online payment gateway integration (BharatKosh/Razorpay)
- SMS/Email notifications (NIC SMS/MSG91 + SendGrid/AWS SES):
 - Application status updates
 - Inspection appointment reminders
 - Certificate issuance alerts
 - Expiry reminders (90, 60, 30 days before)
- Digital signature integration (DSC for LMOs, eSign for GATCs)
- Mobile app for LMOs (offline-first inspection app)
- Advanced search filters (public portal)
- Bulk upload for instruments (Excel import)
- Instrument transfer/ownership change workflow

10.2.2 Deliverables

- Payment gateway integration (tested with sandbox + production)
- SMS/Email gateway integration (tested with real numbers/emails)
- Digital signature integration (tested with test DSC/eSign)
- Mobile app (Android + iOS; inspection workflow only)
- Updated user manuals (include new features)
- API documentation (updated for mobile app)

10.2.3 Timeline

Week	Tasks
Week 9-10	Payment gateway integration (backend + frontend)
Week 11-12	SMS/Email gateway integration (templates, triggers)
Week 13	Digital signature integration (DSC/eSign APIs)
Week 14-15	Mobile app development (LMO inspection workflow)
Week 16	Integration testing, UAT, deployment to production

10.3 Phase 3: Advanced Features - Weeks 17-24

10.3.1 Scope

Goal: Add advanced analytics, enforcement features, and scalability improvements.

Features:

- Advanced MIS & analytics (Elasticsearch + Kibana / Power BI):
 - Compliance heatmaps (by district, instrument type)
 - LMO/GATC performance dashboards
 - Revenue collection reports
 - Public verification statistics
- Enforcement module enhancements:
 - Flying squad inspections (random QR scans)
 - Penalty calculation & compounding (Section 27)
 - Seizure auction management
- Blockchain pilot (optional):
 - Certificate hash stored on government blockchain (NIC Chain)
 - Immutable audit trail
- API integrations:

- GSTN (verify GSTIN during registration)
- UIDAI (Aadhaar OTP for login)
- Transport Ministry (for vehicle weighbridges)
- Petroleum Ministry (for fuel dispensers)
- Multi-language support (Hindi + state languages)
- Accessibility improvements (WCAG 2.1 AA compliance)

10.3.2 Deliverables

- Analytics dashboards (configured in Kibana/Power BI)
- Enforcement module (flying squad app + penalty workflow)
- Blockchain integration (if approved by ministry)
- API integrations (GSTN, UIDAI, etc.)
- Multi-language UI (Hindi + 2-3 state languages)
- Accessibility audit report (WCAG 2.1 AA)

10.3.3 Timeline

Week	Tasks
Week 17-18	Analytics setup (Elasticsearch + Kibana / Power BI)
Week 19-20	Enforcement module development
Week 21	Blockchain pilot (if approved)
Week 22-23	API integrations (GSTN, UIDAI, etc.)
Week 24	Multi-language support, accessibility improvements, final UAT

10.4 Phase 4: Scale & Optimize - Weeks 25+

10.4.1 Scope

Goal: Optimize for scale, add AI/ML features, and expand to all states.

Features:

- Performance optimization:
 - Database indexing & query optimization
 - Caching layer (Redis)
 - CDN for static assets (certificate PDFs, photos)
- AI/ML features (optional):
 - OCR for TAC/document verification

- Anomaly detection (fake certificates, unusual patterns)
- Predictive analytics (instruments likely to fail inspection)
- Nationwide rollout:
 - Onboard all states/UTs
 - Migrate existing state portals (e-Mapan Punjab, Karnataka, etc.)
 - Training for all LMOs/GATCs/ACs
- Continuous improvement:
 - User feedback collection (in-app surveys)
 - Regular feature updates (quarterly releases)
 - Security audits (annual penetration testing)

10.4.2 Deliverables

- Performance benchmark report (response times, throughput)
- AI/ML models (if implemented)
- State onboarding plan (timeline, training schedule)
- User feedback report (quarterly)
- Security audit report (annual)

11. API & Integration Requirements

11.1 Internal APIs (For Frontend)

11.1.1 Authentication APIs

- POST `/api/auth/register` (owner registration)
- POST `/api/auth/login` (all roles)
- POST `/api/auth/logout` (all roles)
- POST `/api/auth/forgot-password` (password reset request)
- POST `/api/auth/reset-password` (password reset with token)
- POST `/api/auth/change-password` (logged-in user)

11.1.2 Instrument APIs

- POST `/api/instruments/register` (pre-registration)
- GET `/api/instruments/list` (owner's instruments)
- GET `/api/instruments/:id/passport` (instrument passport)
- PUT `/api/instruments/:id/decommission` (decommission instrument)

11.1.3 Application APIs

- POST /api/applications/submit (new application)
- GET /api/applications/list (owner's applications)
- GET /api/applications/:id/status (track status)
- POST /api/applications/:id/upload-receipt (payment receipt upload)
- PUT /api/applications/:id/scrutinize (LMO scrutiny: accept/clarification/reject)
- PUT /api/applications/:id/schedule (LMO/GATC scheduling)

11.1.4 Inspection APIs

- POST /api/inspections/submit (LMO field inspection)
- POST /api/inspections/gatc-submit (GATC test report)
- PUT /api/inspections/:id/approve (LMO approve GATC report)
- GET /api/inspections/:id/report (download inspection report PDF)

11.1.5 Certificate APIs

- POST /api/certificates/generate (LMO triggers generation)
- GET /api/certificates/:id/download (download PDF)
- GET /api/certificates/list (owner's certificates)

11.1.6 Public Verification APIs

- GET /api/public/verify/:certificateNumber (search by certificate number)
- GET /api/public/verify-by-serial/:serialNumber (search by instrument serial)
- GET /api/public/verify-by-name/:ownerName (search by owner name)
- POST /api/public/report-discrepancy (whistleblower report)

11.2 External APIs (For Future Integrations)

11.2.1 Payment Gateway (Phase 2)

- **BharatKosh** (government payment gateway)
 - POST /payment/initiate (create payment session)
 - GET /payment/status/:transactionId (check payment status)
 - POST /payment/refund (refund payment; if needed)
- **Razorpay/PayU** (private gateways)
 - Similar endpoints for payment initiation, status, refund

11.2.2 SMS Gateway (Phase 2)

- **NIC SMS Gateway (government)**
 - POST /sms/send (send SMS to mobile number)
 - GET /sms/status/:messageId (check delivery status)
- **MSG91/TextLocal (private)**
 - Similar endpoints for sending SMS, checking status

11.2.3 Email Gateway (Phase 2)

- **SendGrid/AWS SES (private)**
 - POST /email/send (send email with template)
 - GET /email/status/:messageId (check delivery status)

11.2.4 Digital Signature (Phase 2)

- **UIDAI eSign**
 - POST /esign/generate-otp (send OTP to Aadhaar-linked mobile)
 - POST /esign/sign (sign document with OTP)
- **NIC DSC Service**
 - POST /dsc/sign (sign document with DSC token)

11.2.5 GSTN (Phase 3)

- **GST API (government)**
 - GET /gst/verify/:gstIn (verify GSTIN validity, fetch business name)

11.2.6 UIDAI Aadhaar (Phase 3)

- **Aadhaar OTP API**
 - POST /aadhaar/send-otp (send OTP to Aadhaar-linked mobile)
 - POST /aadhaar/verify-otp (verify OTP, fetch name/address)

11.3 File Storage Requirements

11.3.1 Storage Backend

- **Option 1:** Local file system (MVP; simple deployment)
- **Option 2:** Cloud storage (AWS S3, Azure Blob, NIC Cloud Storage) for scalability

11.3.2 File Organization

```
/uploads
/users
 /{userId}
  - profile_photo.jpg
  - gst_certificate.pdf
  - trade_license.pdf
/instruments
 /{instrumentId}
  - tac_certificate.pdf
  - photo_1.jpg
  - photo_2.jpg
/applications
 /{applicationId}
  - payment_receipt.pdf
  - purchase_invoice.pdf
  - previous_certificate.pdf
/inspections
 /{applicationId}
  - inspection_report.pdf
  - photo_display.jpg
  - photo_seal.jpg
  - raw_data.csv (GATC)
/certificates
 /{certificateId}
  - certificate.pdf
/grievances
 /{ticketId}
  - complaint_photo.jpg
  - resolution_notes.pdf
```

11.3.3 File Size Limits

- Images (JPG/PNG): Maximum 5MB per file
- Documents (PDF): Maximum 10MB per file
- Raw data (CSV/Excel): Maximum 20MB per file
- Total per application: Maximum 50MB (all files combined)

12. Testing Requirements

12.1 Unit Testing

12.1.1 Coverage Requirements

- Minimum 80% code coverage for backend APIs
- Minimum 70% code coverage for frontend components
- Critical paths must have 100% coverage:

- Fee calculation logic
- MPE calculation logic
- Certificate generation logic
- GATC routing logic

12.1.2 Test Cases (Examples)

- **Fee Calculation:**
 - Test base fee for different instrument types/capacities
 - Test late fee calculation (0, 1, 2, 3 quarters delayed)
 - Test grace period (re-verification within same quarter → no late fee)
- **MPE Calculation:**
 - Test MPE limits for different instrument classes (I, II, III, IIII)
 - Test pass/fail logic (observed error vs. MPE)
- **GATC Routing:**
 - Test routing for GATC-approved instruments (must route to GATC)
 - Test routing for non-GATC instruments (must route to LMO)
- **Validity Period:**
 - Test "Valid Until" calculation for different instrument types
 - Test quarter system (re-verification within quarter → no late fee)

12.2 Integration Testing

12.2.1 End-to-End Flows

- **Owner Registration → Instrument Registration → Application Submission → Payment Upload → LMO Scrutiny → Inspection → Certificate Issuance → Public Verification**
 - Test complete flow with valid data
 - Test with invalid data (validation errors at each step)
- **GATC Flow:**
 - Application → GATC selection → Test report upload → LMO approval → Certificate issuance
- **Rejection Flow:**
 - Application → LMO scrutiny → Rejection → Appeal → Appeal decision → Re-approval
- **Renewal Flow:**
 - Expiring instrument → Renewal application → Late fee calculation → Inspection → New certificate

12.2.2 Role-Based Access Testing

- Test that each role can only access authorized endpoints:
 - Owner cannot access LMO endpoints
 - LMO cannot access other district's applications
 - GATC cannot issue certificates (only recommend)
 - Public cannot access application status (only certificate summary)

12.3 Security Testing

12.3.1 Penetration Testing

- SQL injection testing (all input fields)
- XSS (Cross-Site Scripting) testing (all user inputs displayed in UI)
- CSRF (Cross-Site Request Forgery) testing (all state-changing endpoints)
- Authentication bypass testing (try accessing endpoints without login)
- Authorization testing (try accessing other user's data)

12.3.2 Data Validation Testing

- Test all file uploads (reject invalid file types, oversized files)
- Test all numeric inputs (reject negative numbers, non-numeric characters)
- Test all date inputs (reject past dates where not allowed, future dates beyond reasonable range)
- Test all dropdowns (reject values not in list)

12.4 Performance Testing

12.4.1 Load Testing

- Simulate 1000 concurrent users (owners submitting applications)
- Simulate 100 concurrent LMOs (uploading inspection reports)
- Simulate 5000 concurrent public users (verifying certificates)
- Measure response times (target: < 2 seconds for all endpoints)
- Measure throughput (target: 100 requests/second)

12.4.2 Stress Testing

- Simulate 10,000 concurrent users (peak load)
- Measure system behavior (graceful degradation vs. crash)
- Identify bottlenecks (database queries, file uploads, certificate generation)

12.4.3 Database Performance

- Test query performance for large datasets (1 million instruments, 500,000 applications)
- Optimize slow queries (add indexes, rewrite queries)
- Test pagination (large lists: instruments, applications, certificates)

12.5 User Acceptance Testing (UAT)

12.5.1 Test Scenarios (Real-World)

- **Scenario 1:** Kirana store owner registers 5 weighing scales, applies for verification, receives certificates
- **Scenario 2:** Hospital registers baby weighing scale (GATC instrument), sends to GATC, receives certificate
- **Scenario 3:** Fuel station owner applies for re-verification of 10 fuel dispensers (expired 2 months ago; late fee applies)
- **Scenario 4:** Consumer scans QR code on certificate at shop, verifies validity, reports discrepancy (fake certificate)
- **Scenario 5:** LMO conducts 20 inspections in a day, issues 15 certificates, rejects 5 instruments
- **Scenario 6:** Assistant Controller reviews 10 seizure proposals, approves 7, rejects 3

12.5.2 UAT Participants

- 5-10 Instrument Owners (real traders, businesses)
- 5-10 LMOs (from District Legal Metrology Offices)
- 2-3 GATCs (accredited labs)
- 2-3 Assistant Controllers (district supervisors)
- 2 Central Administrators (DoCA officials)

12.5.3 UAT Duration

- 2 weeks (Phase 1 MVP)
- 1 week (Phase 2 Enhanced)
- 1 week (Phase 3 Advanced)

13. Deployment & Maintenance Requirements

13.1 Deployment Environment

13.1.1 Server Requirements (MVP)

- **Application Server:**
 - CPU: 4 cores (minimum)
 - RAM: 8 GB (minimum)
 - Storage: 100 GB SSD (for application + file uploads)
 - OS: Ubuntu 20.04 LTS / CentOS 8 / Windows Server 2019
- **Database Server:**
 - CPU: 4 cores (minimum)
 - RAM: 8 GB (minimum)
 - Storage: 200 GB SSD (for database + indexes)
 - OS: Same as application server
- **File Storage:**
 - Option 1: Local storage on application server (100 GB)
 - Option 2: Cloud storage (AWS S3, Azure Blob; scalable)

13.1.2 Network Requirements

- **Bandwidth:** 100 Mbps (minimum)
- **SSL Certificate:** Required (HTTPS for all traffic)
- **Firewall:** Restrict database access to application server only
- **Load Balancer:** Optional for MVP (required for Phase 3+ with high traffic)

13.1.3 Domain & DNS

- **Domain:** `verify.emaap.gov.in` (or state-specific subdomain)
- **DNS:** Point to application server IP
- **SSL:** Let's Encrypt (free) or government CA certificate

13.2 Deployment Process

13.2.1 Staging Environment

- Deploy to staging server first
- Run all test cases (unit, integration, security, performance)
- UAT with real users (LMOs, owners, GATCs)
- Fix bugs identified in UAT
- Get sign-off from stakeholders (Assistant Controller, Central Admin)

13.2.2 Production Deployment

- Deploy to production server (during low-traffic hours: 2 AM - 6 AM)
- Run smoke tests (critical flows: registration, application, certificate generation)
- Monitor logs for errors (first 24 hours)
- Notify users via dashboard announcement (if downtime expected)

13.2.3 Rollback Plan

- Keep previous version deployed on staging server
- If critical bug found in production → rollback to previous version within 1 hour
- Document rollback reason, fix bug, redeploy after testing

13.3 Monitoring & Maintenance

13.3.1 Application Monitoring

- **Uptime Monitoring:** Ping application every 5 minutes (uptime target: 99.5%)
- **Error Monitoring:** Log all errors (5xx responses) → alert on-call developer
- **Performance Monitoring:** Track response times (alert if > 5 seconds for any endpoint)
- **Database Monitoring:** Track query performance, connection pool usage, disk space

13.3.2 Log Management

- **Application Logs:** Store all application logs (info, warn, error)
- **Audit Logs:** Store all audit logs (immutable; separate table)
- **Access Logs:** Store all HTTP access logs (for security analysis)
- **Log Retention:** 90 days (online), 10 years (archived)

13.3.3 Backup & Recovery

- **Database Backup:** Daily full backup (retained for 30 days)
- **File Backup:** Daily incremental backup (retained for 30 days)
- **Disaster Recovery:** Restore from backup within 4 hours (RTO: 4 hours, RPO: 24 hours)
- **Backup Testing:** Test restore procedure monthly (ensure backups are valid)

13.3.4 Security Patches

- **OS Patches:** Apply security patches monthly (Ubuntu/CentOS updates)
- **Application Patches:** Update dependencies (npm/pip packages) monthly
- **Vulnerability Scanning:** Run automated scan weekly (OWASP ZAP, Nessus)
- **Penetration Testing:** Annual third-party penetration test

13.4 Support & Helpdesk

13.4.1 User Support Channels

- **Email Support:** support-lm@emaap.gov.in (response within 24 hours)
- **Phone Support:** Toll-free number 1800-11-4000 (during office hours: 9 AM - 6 PM)
- **In-App Help:** FAQs, user manuals (PDF), video tutorials (links to YouTube)
- **Grievance Module:** For application-specific issues (tracked via ticket number)

13.4.2 Support SLA

- **Critical Issues** (system down, data loss): Response within 1 hour, resolution within 4 hours
- **High Priority** (feature broken, data incorrect): Response within 4 hours, resolution within 24 hours
- **Medium Priority** (UI bug, minor feature request): Response within 24 hours, resolution within 3 days
- **Low Priority** (cosmetic issue, enhancement): Response within 3 days, resolution in next release

13.4.3 Training Materials

- **User Manuals:** PDF for each role (owner, LMO, GATC, AC, admin)
- **Video Tutorials:** 5-10 minute videos for each major workflow (upload to YouTube)
- **FAQs:** Top 50 questions (updated quarterly based on support tickets)
- **Webinars:** Monthly training sessions for new LMOs/GATCs (recorded for future reference)

14. Appendices

14.1 Appendix A: Database Schema (ER Diagram Description)

14.1.1 Tables

users

- id (PK)
- user_id (unique: LM-USER-YYYY-XXXXXX)
- role (enum: owner, lmo, gatc, assistant_controller, central_admin)
- business_name (nullable; for owners)
- gstin (nullable; for owners)
- pan (nullable; for owners)
- aadhaar (nullable; encrypted)
- mobile (nullable)
- email (nullable)
- password_hash (encrypted)
- jurisdiction_pincode_range (nullable; for LMOs)
- accreditation_number (nullable; for GATCs)
- accreditation_categories (nullable; for GATCs; JSON array)
- accreditation_validity (nullable; for GATCs)
- assigned_districts (nullable; for ACs; JSON array)
- status (enum: active, inactive, locked)
- created_at (timestamp)
- updated_at (timestamp)

instruments

- id (PK)
- instrument_id (unique: IN-MET-YYYY-XXXXXX)
- owner_id (FK → users.id)
- type (enum: electronic_scale, platform_scale, weighbridge, fuel_dispenser, tank_lorry, analytical_balance, flow_meter, etc.)
- make_model (text)
- serial_number (text)
- capacity (numeric)
- division (numeric)
- manufacturing_year (integer)

- tac_certificate_url (nullable; file path)
- installation_address (text)
- installation_pincode (text)
- status (enum: registered, under_verification, active, expiring_soon, expired, decommissioned, seized)
- created_at (timestamp)
- updated_at (timestamp)

applications

- id (PK)
- application_id (unique: APP-LM-YYYY-XXXXXX)
- instrument_id (FK → instruments.id)
- owner_id (FK → users.id)
- verification_type (enum: initial, re_verification, after_repair)
- preferred_mode (enum: lmo_field, gatc_lab)
- preferred_date (date)
- preferred_time_slot (enum: 9-11, 11-1, 2-4, 4-6)
- calculated_fee (numeric)
- late_fee (numeric; nullable)
- total_fee (numeric)
- payment_receipt_url (nullable; file path)
- payment_status (enum: pending, uploaded, verified, rejected)
- status (enum: draft, submitted, payment_uploaded, payment_verified, under_scrutiny, assigned, inspection_scheduled, inspection_completed, certificate_under_issuance, certificate_issued, rejected, appeal_filed, appeal_decision)
- assigned_lmo_id (FK → users.id; nullable)
- assigned_gatc_id (FK → users.id; nullable)
- scrutiny_remarks (nullable; text)
- created_at (timestamp)
- updated_at (timestamp)

inspections

- id (PK)
- application_id (FK → applications.id)
- inspector_id (FK → users.id; LMO or GATC)
- inspection_type (enum: lmo_field, gatc_lab)
- inspection_date (date)

- physical_inspection_data (JSON; checklist results)
- metrological_test_data (JSON; test results, observed values, errors, MPE limits)
- overall_result (enum: pass, fail)
- seal_number (nullable; text)
- gps_coordinates (nullable; JSON: latitude, longitude)
- photos_urls (JSON array; file paths)
- inspection_report_url (file path; PDF)
- gatc_test_report_url (nullable; file path; PDF for GATC)
- gatc_recommendation (nullable; enum: certify, reject; for GATC only)
- lmo_approval_status (nullable; enum: approved, clarification_asked, rejected; for GATC reports)
- lmo_approval_remarks (nullable; text)
- created_at (timestamp)
- updated_at (timestamp)

certificates

- id (PK)
- certificate_number (unique: LM/VER/YYYY/XXXXX)
- application_id (FK → applications.id)
- instrument_id (FK → instruments.id)
- owner_id (FK → users.id)
- issue_date (date)
- valid_until (date)
- certificate_pdf_url (file path)
- qr_code_data (text; encoded data)
- seal_number (text)
- stamp_quarter (text; e.g., Q3-2026)
- lmo_id (FK → users.id; who issued)
- gatc_id (FK → users.id; nullable; if GATC tested)
- status (enum: active, expired, superseded, revoked)
- superseded_by (FK → certificates.id; nullable; if re-issued)
- created_at (timestamp)

grievances

- id (PK)
- ticket_number (unique: GRV-LM-YYYY-XXXXX)

- applicant_id (FK → users.id)
- application_id (FK → applications.id; nullable)
- certificate_id (FK → certificates.id; nullable; for public discrepancy reports)
- grievance_type (enum: lmo_no_visit, gatc_damaged_instrument, certificate_error, payment_dispute, fake_certificate, expired_in_use, seal_damaged, other)
- description (text)
- photos_urls (JSON array; nullable)
- assigned_to_id (FK → users.id; LMO/GATC/AC)
- resolution_notes (nullable; text)
- resolution_documents_url (nullable; file path)
- status (enum: pending, in_progress, resolved, escalated)
- created_at (timestamp)
- updated_at (timestamp)

seizures

- id (PK)
- seizure_id (unique: SEZ-LM-YYYY-XXXXX)
- instrument_id (FK → instruments.id)
- owner_id (FK → users.id)
- lmo_id (FK → users.id; who proposed)
- reason (enum: fake_certificate, expired_certificate, tampered_seal, accuracy_failure, other)
- seizure_location (text)
- seizure_date (datetime)
- photos_urls (JSON array)
- witness_details (nullable; JSON: name, contact)
- status (enum: proposed, approved, rejected, additional_info_asked, disposed)
- disposal_method (nullable; enum: returned, auctioned, destroyed)
- disposal_date (nullable; date)
- disposal_amount (nullable; numeric; if auctioned)
- ac_approval_id (FK → users.id; Assistant Controller who approved)
- ac_approval_remarks (nullable; text)
- created_at (timestamp)
- updated_at (timestamp)

audit_logs

- id (PK)

- user_id (FK → users.id)
- action (text; e.g., "LOGIN", "APPLICATION_SUBMIT", "CERTIFICATE_ISSUE")
- entity_type (text; e.g., "application", "certificate", "instrument")
- entity_id (integer; ID of affected entity)
- old_value (nullable; JSON; previous state)
- new_value (nullable; JSON; new state)
- ip_address (text)
- created_at (timestamp)

14.2 Appendix B: Sample Data (For Testing)

14.2.1 Sample Instrument Types

```
[
  {"type": "electronic_scale", "capacity_range": "0-15kg", "base_fee": 50, "validity_months": 12},
  {"type": "electronic_scale", "capacity_range": "15-100kg", "base_fee": 100, "validity_months": 12},
  {"type": "weighbridge", "capacity_range": "0-50t", "base_fee": 300, "validity_months": 12},
  {"type": "analytical_balance", "capacity_range": "any", "base_fee": 2000, "validity_months": 12},
  {"type": "flow_meter", "capacity_range": "any", "base_fee": 3000, "validity_months": 12}
]
```

14.2.2 Sample Users (For Testing)

```
[
  {"user_id": "LM-USER-2026-00001", "role": "owner", "business_name": "Mumbai Kirana", "jurisdiction_pincode_range": "400001-400005"},
  {"user_id": "LM-USER-2026-00002", "role": "lmo", "jurisdiction_pincode_range": "400001-400005"},
  {"user_id": "LM-USER-2026-00003", "role": "gadc", "accreditation_number": "GATC-2026-00001"},
  {"user_id": "LM-USER-2026-00004", "role": "assistant_controller", "assigned_districts": ["Mumbai", "Thane"]},
  {"user_id": "LM-USER-2026-00005", "role": "central_admin", "assigned_districts": []}
]
```

14.3 Appendix C: Glossary

Term	Definition
LMO	Legal Metrology Officer (field inspector)
GATC	Government Approved Test Centre (accredited laboratory)
TAC	Type Approval Certificate (from manufacturer)
MPE	Maximum Permissible Error (tolerance limit for instruments)
OIML	International Organization of Legal Metrology (sets international standards)
Schedule VIII	Certificate format prescribed in Legal Metrology (General) Rules, 2011

Term	Definition
Rule 13	Validity periods for instruments (Legal Metrology (General) Rules, 2011)
Rule 14	Fee structure for verification (Legal Metrology (General) Rules, 2011)
Section 24	Verification and stamping requirement (Legal Metrology Act, 2009)
Section 15	Inspection and seizure powers (Legal Metrology Act, 2009)
Instrument Passport	Complete lifecycle history of an instrument (all certificates, inspections, seals)
Quarter System	Re-verification within same quarter as expiry → no late fee (Q1: Jan-Mar, Q2: Apr-Jun, etc.)

14.4 Appendix D: References

1. Legal Metrology Act, 2009: <https://consumeraffairs.gov.in/pages/legal-metrology-act>
2. Legal Metrology (General) Rules, 2011: <https://www.indiacode.nic.in/handle/123456789/1362>
3. Legal Metrology (GATC) Rules, 2013: https://consumeraffairs.gov.in/sites/default/files/9-lm_gatc_rules_2013.pdf
4. eMaap Portal: <https://emaap.gov.in/>
5. DigiPass Reference: <https://digipassgov.vercel.app/>
6. OIML R76 Standard: <https://www.oiml.org/en/publications/r-r076-e00.pdf>

Document Version: 1.0

Last Updated: September 24, 2026

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This documentation is complete and ready for developers to build the full-stack application. All requirements, flows, roles, business rules, and implementation phases are covered in detail.